Class - X

Physics

Chapter - I

ELECTRICITY

Electricity plays an important role in our modern times.
Electricity is used in our homes, industri and in transport.

★ Highlight of the chapter

- Electric potential
- Electric current
- Electric power
- Resistance and Resistivity
- Heating effect of electric current.

In lower classes we studied about the electric charges. Let discuss it first before going through the chapter.

If we bring a plastic comb near some very tiny piece of paper, it will not have any effect on them. However if we rubbed the comb and brought near the piece of paper we find that the comb now attract the piece of paper towards itself.

Similarly, a glass rod rubbed with silk

cloth and an ebonite rod rubbed with woolen cloth also acquire the ability to attract piece of paper. These observation comes to a very conclusion that their are two types of charges one is called Positive charge and another is called negative charge.

★ Important properties of electric charge is that:

- opposite charges (or Unlike charges) attract each other.
- similar charges (or like charges) repel each other.



Interaction between charges

Charge: The charge is an intrinsic property of matter by virtue of which it can exert electromagnetic Force.

SI unit of electric charge is Coulomb (C)

★ One coulomb is that quantity of electric charge which exert a force of 9×10^9 newtons on an equal charge placed at a distance of 1 metre from it.

Charges on the matter

- Proton (+ve charge) → 1.6×10^{-19} C
- Electron (-ve charge) → 1.6×10^{-19} C
- Neutron (Neutral)

Note It is a case of single neutral atom

Now take an example to understand it clearly.

Q: Calculate the number of electrons constituting one coulomb of charge.

Sol^n:- We know that

Amount of charge is 1.6×10^{-19} C

No. of electron = 1

If charge is 1C, then No. of electron $= \frac{1}{1.6 \times 10^{-19}} \times 1$

$= \frac{10^{19}}{1.6}$

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$$= \frac{10}{1.6} \times 10^{18}$$
$$= 6.25 \times 10^{18}$$

Thus, 6.25×10^{18} electron taken together constitute 1 coulomb of charge.

★ Conductors and Insulators

Conductors :- The substances through which electricity can flow are called conductors.
eg:- silver, copper, gold, aluminium etc

Insulators :- The substances through which electricity cannot flow are called Insulators.
eg:- dry wood, plastic, glass, dry air rubber etc.

Note The presence of free electron in a substance makes it a conductor.

Electric Potential

The electric potential at a point in an electric field is defined as the work done in moving a unit positive charge from infinity to that point

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The SI unit of electric potential difference is volt (V).

If 'W' is the work done 'q' is the charge then electric potential $V = W/q$

Potential Difference

The potential difference between two points is the work done in carrying a unit positive charge from one point to the other.

$$\text{Potential difference, } V = \frac{\text{Work done}}{\text{Quantity of charge moved}}$$

$$\boxed{V = \frac{W}{q}}$$

The potential difference between two points is said to be 1 Volt if 1 Joule of work is done in moving 1 coulomb of electric charge from one point to the other.

$$1 \text{ volt} = \frac{1 \text{ joule}}{1 \text{ coulomb}}$$

$$1V = \frac{1J}{1C}$$

$$\boxed{1V = 1JC^{-1}}$$

NCERT

Q: How much energy is given to each coulomb of charge passing through a 6V battery?

Soln:- The term 'each coloumb' means 'every one coloumb'

So the charge here is 1C.
The potential diffence (V) = 6V

$$\text{Potential difference} = \frac{\text{Work done}}{\text{charge moved}}$$

or, $$V = \frac{W}{q}$$

$$6 = \frac{W}{1}$$

$$W = 6 \times 1$$

Work done, W = 6 Joules

Voltmeter

It is a device used to measure potential difference across the two points in a circuit.

- It is always connected in parallel across the points where potential difference is to be measured.
- A voltmeter has a high resistance.

Note: It is connected in parallel so to measure accurate amount of energy used between the two ends and amount of current through it become very less so the current through the circuit is not altered.

ELECTRIC CURRENT

Flow of electric charges (called electrons) in a conductor is called Electric current.

The magnitude of electric current in a conductor is the amount of electric charge passing through a given point of the conductor in one seconds

$$\text{Current}, I = \frac{q}{t}$$

The SI unit of electric current is Ampere

Q: Define 1 ampere ?

Ans:- When 1 coulomb of charge flow through any cross-section of a conductor in 1 second the electric current flowing through it is said to be 1 ampere.

$$1 \text{ ampere} = \frac{1 \text{ coulomb}}{1 \text{ second}}$$

or, $$1A = \frac{1C}{1s}$$

In the form of "milliampere",

$$1 \text{ milliampere} = \frac{1}{1000} \text{ ampere}$$

or, $$1mA = \frac{1}{1000} A$$

$$\boxed{1mA = 10^{-3} A}$$

In microampere

$$1 \mu A = 10^{-6} A$$

★ Ammeter —+Ⓐ−—

- It is a device used to measure current in the circuit.
- It is always connected in a series in a circuit.
- An ammeter should have very low resistance.

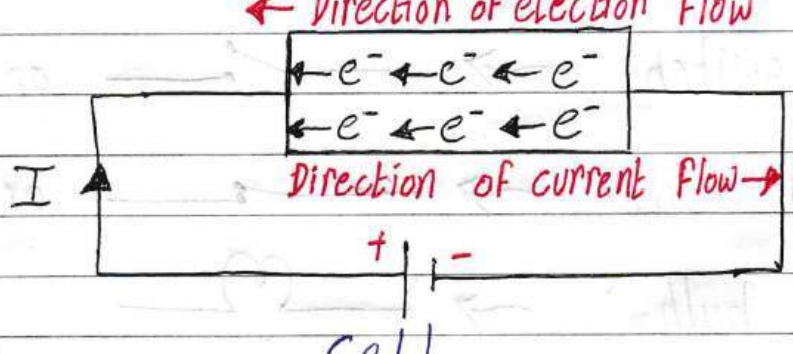
Q: Why should ammeter have low resistance and voltmeter have high resistance?

Ans:- The reason for ammeter resistance to be low is because it can accurately read the current flowing in the circuit.

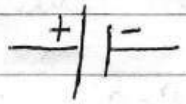
When the ammeter resistance is low, almost all the current in the circuit is allowed to pass through the ammeter. This allow the ammeter to accurately read the current in the circuit. But on the other hand, if the ammeter resistance was more the ammeter would read a value lesser than the actual value flowing through the circuit.

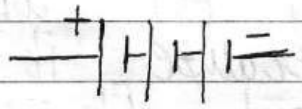
Note:- The conventional direction of electric current is from positive terminal of a cell (or battery) to the negative terminal, through the outer circuit.

The actual flow of electron is however from negative terminal to positive terminal of a cell, which is opposite to the direction of conventional current.



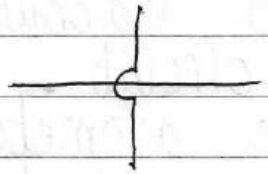
★ Symbols For Electrical components

a.) Cell → 

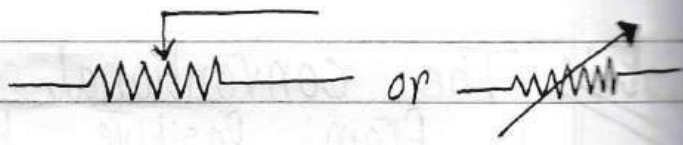
b.) Battery → 

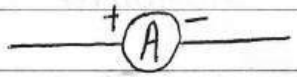
c.) Connecting wire →

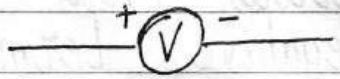
d.) A wire joint → 

e.) Wire crossing without contact → 

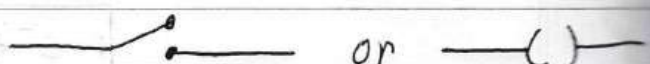
f.) Fixed resistance (or Resistor) →

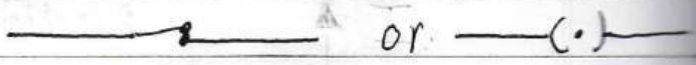
g.) Variable resistance (or Rheostat) → 

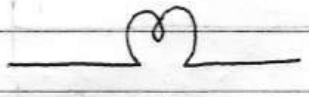
h.) Ammeter → 

i.) Voltmeter → 

j.) Galvanometer → 

k.) An open switch → 

l.) A closed switch → 

m.) Electric bulb → 

Electric Circuit

- A closed-loop path which a current take is called and electric circuit.
- Representation of an electric circuit through symbols is called circuit diagram

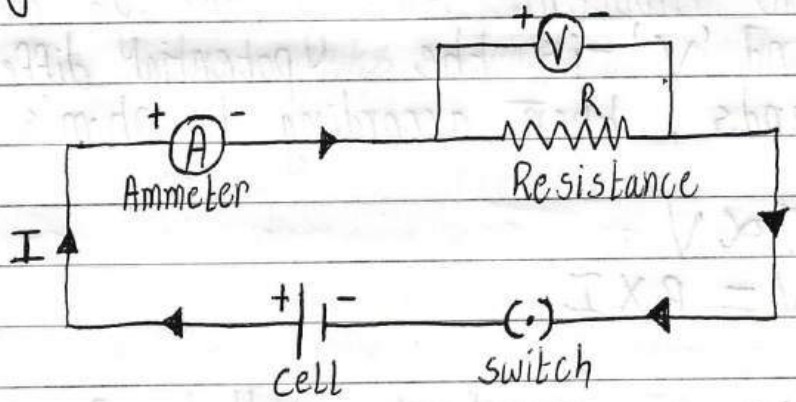


Fig :- Circuit diagram

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★ OHM'S LAW

At constant temperature, the current flowing through a conductor is directly proportional to the potential difference across its ends.

If 'I' is the current flowing through a conductor and 'V' is the potential difference across its ends, then according to Ohm's Law:-

$$I \propto V$$

or $$V = R \times I$$

Where R is a constant called Resistance of the conductor.

The above equation can also be written as:

$$\boxed{\frac{V}{I} = R}$$

The above equation is a mathematical expression of Ohm's Law.

In words.

$$\frac{\text{Potential diff}}{\text{current}} = \text{Constant (Resistance)}$$

The ratio of Potential difference applied between the ends of a conductor and the current flowing through it is a constant quantity called resistance.

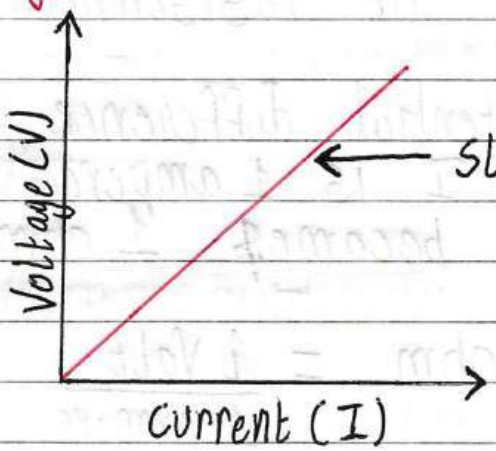
From the relation we deduce that.

i) the current is directly proportional to potential difference.

ii) the current is inversely proportional to resistance.

i.e, If the potential difference across the ends of a conductor is doubled, the current through it also gets doubled and vice versa.

Voltage - Current characteristics - Ohm's Law



★ The strength of an electric current in a given conductor depends on two factors :-

i) Potential difference

ii) Resistance

Resistance

The property of a conductor due to which it opposes the flow of current through it is called resistance.

$$\text{Resistance} = \frac{\text{Potential difference}}{\text{current}}$$

or $$\boxed{R = \frac{V}{I}}$$

The SI unit of resistance is ohm (Ω)

Now, if the potential difference V is 1 volt and current I is 1 ampere, then Resistance R becomes 1 ohm.

i.e $$1 \text{ ohm} = \frac{1 \text{ volt}}{1 \text{ ampere}}$$

Q: Define 1 ohm ?

Ans:- 1 ohm is the resistance of a conductor such that when a potential difference of 1 volt is applied to its ends, a current of 1 ampere flows through it.

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★ Factors affecting Resistance & How they affect

Resistance is:

- directly proportional to the length of the conductor.
- directly proportional to nature of the conductor.
- directly proportional to the temperature of the conductor.
- inversely proportional to the cross-sectional area of the conductor.

Resistivity

It has been found by experiments that

- The resistance of a given conductor is directly proportional to its length

$$R \propto l$$

- The resistance of a given conductor is inversely proportional to its area of cross-section.

$$R \propto \frac{1}{A}$$

From above relation after combining.

$$R \propto \frac{l}{A}$$

or $\boxed{R = \frac{\rho l}{A}}$

Where ρ (rho) is a constant known as resistivity of the material of the conductor.

We can also write the above equation as

$$\text{Resistivity, } \rho = \frac{RA}{l}$$

$$\text{Unit of resistivity, } \rho = \frac{\text{ohm} \times (\text{metre})^2}{\text{metre}}$$

$$\rho = \text{ohm-metre} \quad (\Omega m)$$

Defination:

The electrical resistance offered by a substance of unit length and unit cross-sectional area is called resistivity.

Note: The resistivity of a substance dosnot depend on its length or thickness. It depends on the nature of the substance and temperature.

★ Resistivity of some common substances at (20°C)

Conductors :

Metals

1. Silver → $1.60 \times 10^{-8}\ \Omega m$
2. Copper → $1.69 \times 10^{-8}\ \Omega m$
3. Aluminium → $2.63 \times 10^{-8}\ \Omega m$
4. Tungsten → $5.20 \times 10^{-8}\ \Omega m$
5. Nickel → $6.84 \times 10^{-8}\ \Omega m$

Alloys

1. Manganin (Cu-Mn-Ni) → $44 \times 10^{-8}\ \Omega m$
2. Constantan (Cu-Ni) → $49 \times 10^{-8}\ \Omega m$
3. Nichrome (Ni-Cr-Mn-Fe) → $110 \times 10^{-8}\ \Omega m$

Semiconductors :

1. Germanium → $0.6\ \Omega m$
2. Silicon → $2300\ \Omega m$

Insulators :

1. Glass → 10^{10} to $10^{14}\ \Omega m$
2. Paper (Dry) → $10^{12}\ \Omega m$
3. Hard rubber → $10^{13} - 10^{16}\ \Omega m$

★ Combination of Resistance

The resistance can be combined in two ways

i) In series
ii) In parallel

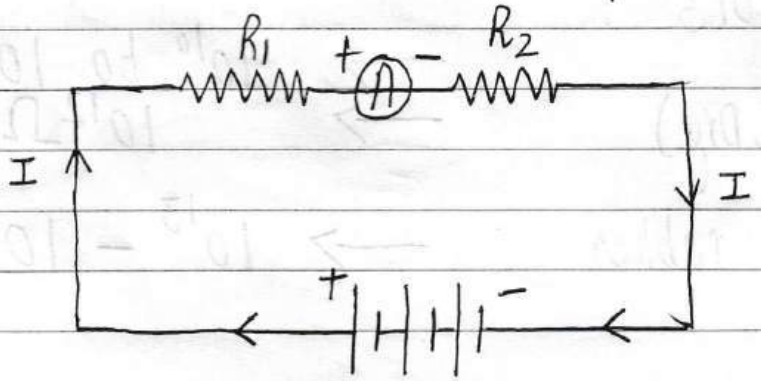
Resistance in Series

When two or more resistances are connected end to end consecutively they are said to be connected in series.

In series combination of resistances, the equivalent resistance is equal to the sum of individual resistances.

Thus the equivalent resistance is always greater than the individual resistance.

The resistance shown below is connected in series.



We can see that the resistances in series combination are connected end to end. The equivalent resistance (R_{eq}) can be given as,

$$R_{eq} = R_1 + R_2$$

When resistances are connected in series, the current passing through them is same but they can have different potential diffence across their ends.

Resistance in parallel

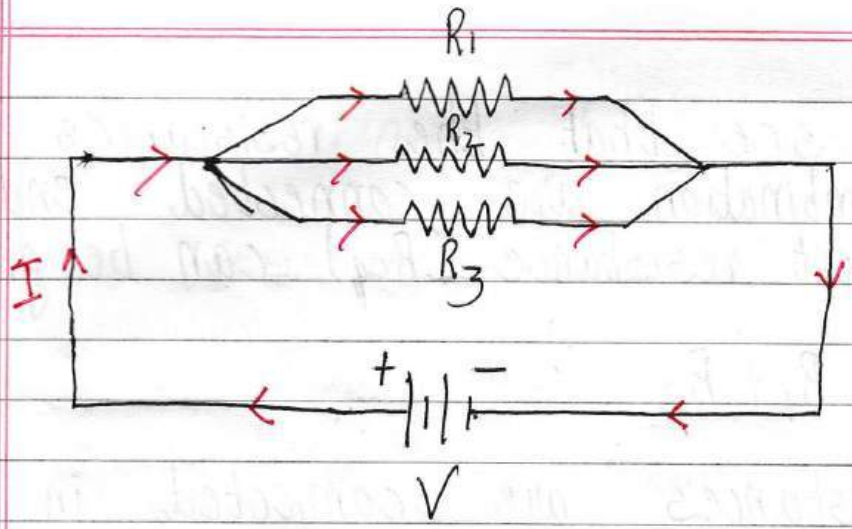
When two or more resistances are connected between the same two points they are said to be connected in parallel.

In parallel combination of resistance, the reciprocal of equivalent resistance is equal to the sum of reciprocals of individual resistances.

Thus the equivalent resistance is always smaller than the individual resistance.

Take a look at resistance connected in parallel.

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We can see that the resistances in parallel combination are connected such that their ends connected to a common point.

The equivalent resistance (R_{eq}) can be given as,

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

When resistances are connected in parallel, the potential difference across their end points is same but they can have different current passing through them.

Domestic Electric Circuits

In general domestic Electric circuit is a parallel combination / connection.

To understand it Lets see the disadvantage of series connection

Disadvantages of series circuit for domestic Wiring

1. In series circuit, if one electrical appliance stop working due to some defect, then all of the appliances also stop working.
2. In this all the electrical appliances have only one switch due to which they cannot be turned ON or OFF separately.
3. In series circuit, the appliances donot get the same voltage (220V) as that of the power supply Line.
4. In this the overall resistance of the circuit increases too much due to which the current from the power supply is Low.

Advantages of Parallel circuits in Domestic

1. In parallel circuits if one electrical appliances stops working due to some defect then all other appliances keep working normally.

2. In this each electrical appliances has its own switch due to which it can be turned ON or OFF independently.

3. In this each appliances gets the same voltage 220V

4. In this the overall resistance of the household circuit is reduced due to which the current from the power supply is high.

ELECTRIC POWER

We know that the rate of doing work is called power, so electric power is the electrical work done per unit time.

$$\text{Power} = \frac{\text{Work done}}{\text{Time taken}}$$

or,

$$P = \frac{W}{T}$$

The SI unit of electric power is Watt (W)

$$1 \text{ Watt} = \frac{1 \text{ Joule}}{1 \text{ second}}$$

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★ Watt is a small unit, therefore a bigger unit of electric power called Kilo watt is used for comercial purposes.

$$1 \text{ Kilowatt} = 1000 \text{ Watt}$$

or

$$1 \text{ kW} = 1000 \text{ W}$$

In other word electric power is defined as electrical energy consumed per second.

Formulae associated with power

We know that

$$P = \frac{W}{t} \qquad \text{— (i)}$$

We have already studied that workdone W by current I when it flows for time t under a potential difference V is given by:-

$$W = V \times I \times t$$

Putting the value of W from (1)

$$P = \frac{V \times I \times t}{t}$$

or $\boxed{P = V \times I}$

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Power P in term of I and R

$$P = V \times I \quad - (i)$$

Now, from ohm's Law we have, $\frac{V}{I} = R$

or $V = IR \quad - (ii)$

Putting the value of V in (i)

$$P = IRI$$

$$\boxed{P = I^2R}$$

Power in term of V and R

$$P = V \times I \quad - (1)$$

From ohm's Law, $\frac{V}{I} = R$

or, $I = \frac{V}{R} \quad - (2)$

Now from (1) and (2)

$$P = V \times \frac{V}{R}$$

$$\boxed{P = \frac{V^2}{R}}$$

★ An Important Formula for calculating Electrical Energy.

Electrical energy in terms of power and time

$$\text{Electric power} = \frac{\text{Work done by electric current}}{\text{Time taken}}$$

Now, according to the Law of conservation of energy

$$\text{Work done by electric current} = \text{Electric energy consumed}$$

So we can write the above equation as

$$\text{Power} = \frac{\text{Electrical energy}}{\text{Time}}$$

$$\text{Electrical energy} = \text{Power} \times \text{Time}$$

$$\boxed{E = P \times t}$$

In words.

The electrical energy consumed by an electrical appliance is given by the Product of its Power rating and Time for which it is used.

Q: Define one watt-hour ?

Ans:- One watt-hour is the amount of electrical energy consumed when an electrical appliance of 1 watt power is used For 1 hour.

Q: Define one Joule of electrical energy ?

Ans One joule is the amount of electrical energy consumed when an appliance of 1 Wattage is used For 1 second.

Q: What is the commercial unit of Electrical energy find its relation with joule.

Ans:- The commercial unit of electrical energy is Kilowatt-hour.

Relation of Kilowatt-hour with Joule

1 Kilowatt-hour = 1 Kilowatt For 1 hour
1 Kilowatt-hour = 1000 watts For 1 hour

But: $1 \text{ watt} = \frac{1 \text{ Joule}}{1 \text{ second}}$

So we can write it as :-

1 Killowatt-hour = 1000 $\frac{\text{joules}}{\text{second}}$ For 1 hour

And, 1 hour = 60 X 60 seconds

so, 1 Kilowatt-hour = 1000 $\frac{\text{joules}}{\text{sec}}$ X 60 X 60 sec

1 Kilowatt-hour = 36,00,000 Joules
= 3.6×10^6 J

Heating Effect of Current

When an electric current is passed through a high resistance wire, like nichrome the resistance wire becomes very hot and produces heat. This is called heating effect of current.

This effect is produced due to transformation of electrical energy into heat energy.

Formula for calculating the heat produced

Since a resistance wire offers resistance to the flow of current, so work must be done by the current continuously to keep itself flowing. We will calculate the work done by a current I when it flows through a resistance R for time t.

Now, when an electric charge Q moves against a potential difference V, the amount of work done is given by:

$$W = Q \times V \quad - (1)$$

We Know,

$$I = \frac{Q}{t}$$

$$Q = I \times t \quad - (2)$$

And from Ohm's Law,

$$V = IR \quad - (3)$$

Now putting, $Q = It$ and $V = IR$ in eqn (1)

$$W = I t I R$$

$$W = I^2 R t$$

Now Assuming that all the electrical work done or all the electrical energy is converted into Heat energy.

Then we can write above equation as

Heat produced, $H = I^2 R t$ Joules

Note:- According to above formula we clearly say that the heat produced in a wire is directly proportional to:

i) Square of current (I^2) [$H \propto I^2$]

ii) Resistance of wire (R) [$H \propto R$]

iii) Time for which the current is passed [$H \propto t$]

Applications of the Heating Effect of Current

1. The heating effect of current is utilised in the working of electrical heating appliances such as electric iron, electric kettle, electric toaster, room heater, geysers etc.
2. It is utilised in electric bulb for producing light.
3. This effect is utilised in electric fuse for protecting household wiring and electrical appliances.